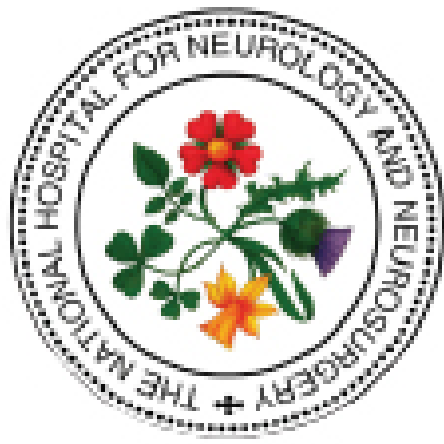




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

LABETALOL HYDROCHLORIDE for RAPID CONTROL of HYPERTENSION

INDICATIONS FOR USE

Severe hypertension when rapid control of blood pressure is essential.

STABILITY AND COMPATIBILITY

See *MEDUSA INJECTABLE MEDICINES GUIDE*

CONTRAINDICATIONS

Contraindicated in second or third degree heart block (unless pacemaker is in situ), cardiogenic shock and other conditions associated with severe and prolonged hypotension or severe bradycardia.

ACUTE REACTIONS

- Bradycardia (excessive bradycardia can be reversed with atropine)
- Heart block, heart failure
- Hypotension, postural hypotension
- Bronchospasm
- Peripheral coldness, paraesthesia
- Hypersensitivity reactions

EXTRAVASATION

- Ideally administered using an infusion pump, via a central venous access device
- Labetalol has a low pH and may cause venous irritation and tissue damage in cases of extravasation.
- If a central venous access device is unavailable, administer via a large peripheral vein monitoring insertion site closely using a recognised phlebitis scoring tool.
- Re-site cannula at first signs of inflammation

INTRAVENOUS LABETALOL as 1st LINE DRUG for RAPID CONTROL of HYPERTENSION

PRESENTATION and ADMINISTRATION

1. 1 ampoule contains labetalol **50mg in 10mL** (5mg/mL)
2. **IV INJECTION:** Give **NEAT as 5mg/mL** solution
3. **IV INFUSION:** Prepare as an **UNDILUTED 5mg/mL** solution preferably via a central venous access device
4. *Labetalol can be diluted to a 1mg/mL solution for both bolus and infusion if necessary to give smaller amounts – preferred diluent is 5% glucose but 0.9% sodium chloride can also be used*

STEP 1

BOLUS DOSE

ONLY USE IF ESSENTIAL TO RAPIDLY REDUCE BP IN HYPERTENSIVE EMERGENCY

- Bolus dose of up to **50mg** can be given by I.V. injection over 1 minute
- Repeat at 5 - 10 minute intervals until satisfactory response
- Total dose should not exceed 200mg in 24 hours (maximum 4 doses)
- Maximum effect usually occurs within 5 minutes and the duration of action is usually about 6 hours

STEP 2

INFUSION

- Start infusion at **4mL/h** (20mg/h)
- Titrate to BP target by adjusting - doubling, maintaining or halving - the infusion as required every 15 to 30 minutes – *allow time to take effect before making adjustments to avoid large swings in BP*
- If BP decreases precipitously, halve the infusion rate or cease depending on severity
- Maximum dose 160mg/h (32mL/h) – *if BP target not reached consider other agent (2nd line: Hydralazine)*

STEP 3

WEANING

- Discontinue by weaning over 1 - 2 hours when clinically appropriate
- Allow each decrease to take effect before making further adjustments to avoid a precipitous increase in BP
- Oral / NG treatment started, IV therapy weaned and stopped within 2-3 days

RAPID CONTROL OF BLOOD PRESSURE IN ACUTE STROKE

ACUTE ISCHAEMIC STROKE (AIS)

POST IV Thrombolysis	<185/110mmHg
POST Thrombectomy +/- IV Thrombolysis	<185/110mmHg
NEITHER IV Thrombolysis nor Thrombectomy	<220/120
NCCU Management	<185mmHg

HAEMORRHAGIC STROKE

<140/90mmHg for at least 7 days

Aim to achieve **target BP within 1 hour** of starting treatment – *while ensuring it does not drop >60mmHg within 1h*

RAPID BP LOWERING should be avoided if elevated ICP suspected, GCS <6 or neurosurgical evacuation pending